E-Commerce Sales Data Analysis Report

Executive Summary:

This report presents a descriptive and exploratory analysis of an E-Commerce sales dataset containing 2000 transactions across multiple regions, product categories, store types, and loyalty segments. The objective was to evaluate overall sales performance, identify top-performing categories and regions, analyze seasonal trends, and derive actionable business insights. The analysis reveals strong regional dominance of the North region, high performance in Sports and Electronics categories, increasing online channel contribution, and strong loyalty engagement particularly among female customers. Seasonal sales peaks occur in Q4 and March–April periods. Strategic recommendations are provided to improve underperforming regions, optimize discount strategies, and enhance customer retention.

Business Objective:

- To evaluate overall sales performance
- To identify top-performing categories and regions
- To analyse monthly sales trends
- To derive business insights for revenue optimization

Dataset: E-Commerce Sales

Data Volume Overview:

1. Total records: 2000
2. Time span: 2023–2025
3. Number of regions: 4
4. Categories: 5
5. Store types: 3

Limitations:

1. October 2025 contains partial data
2. No customer age/income data available
3. Profit margin not included

Analysis Type: Descriptive & Exploratory Analysis

Tools Used: Microsoft Excel

E-Commerce Dataset Attributes:

Attribute Name	Data Type	Variable Type	Description
Sales_ID	Text	Identifier (Primary Key)	Unique transaction identifier

Customer_ID	Text	Identifier	Unique customer identifier
Order_Date	Date	Time Variable	Date on which order was placed
Region	Text	Categorical	Geographic sales region
Gender	Text	Categorical	Customer gender
Category	Text	Categorical	Main product classification
Sub_Category	Text	Categorical	Specific product group under category
Quantity	Integer	Quantitative	Number of units sold
Unit_Price	Currency (Decimal)	Quantitative	Selling price per unit
Total_Amount	Currency (Decimal)	Quantitative	Total revenue per transaction
Discount	Decimal (0–1)	Quantitative	Discount applied on transaction
Payment_Type	Text	Categorical	Mode of payment used
Loyalty_Level	Text	Categorical	Customer loyalty tier

Data Cleaning & Preparation Summary:

- Dataset imported and converted into structured table format
- Data Cleaning process was done in power query
- Missing values were imputed and handled appropriately
- Inconsistent text formats were standardized
- Relevant columns were added to improve analysis quality
- Data types were corrected (dates, currency, numeric)

Visualizations Used:

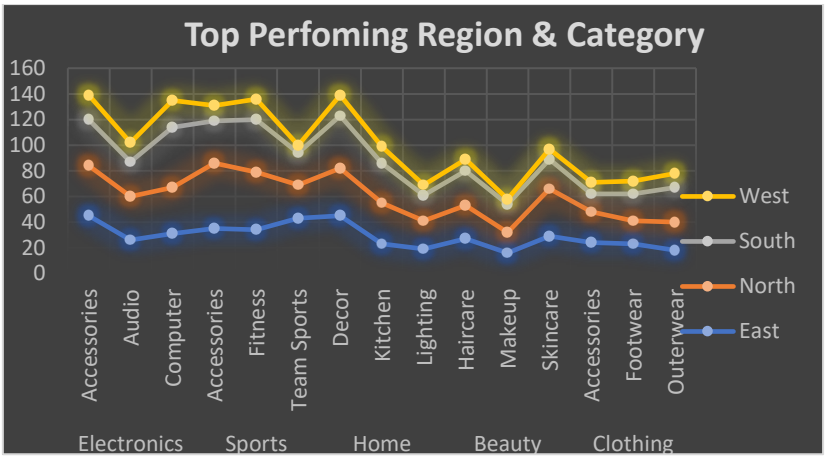
- Line chart for monthly trends
- Clustered column chart for region/category
- Pie chart for payment distribution
- Stacked bar for loyalty analysis

Analytical Techniques Applied:

- Pivot Tables
- Trend Analysis
- Percentage Contribution Analysis
- Comparative Analysis

Key Findings:

1)Top Performing Sales Category & Region:



STRATEGIC INSIGHTS:

- North region contributes the highest share of total sales (approx. 31%), significantly outperforming West..
- West significantly underperforms across all categories.
- Electronics, Sports, and Home are top-performing categories.
- Clothing shows lowest demand overall.

💼 BUSINESS RECOMMENDATIONS:

- ✓ Increase marketing investment in West region.
- ✓ Expand Electronics & Sports inventory in North and South.
- ✓ Promote Clothing category with seasonal offers.
- ✓ Focus product bundling in Decor and Computer segments.

2)Sales Performance:

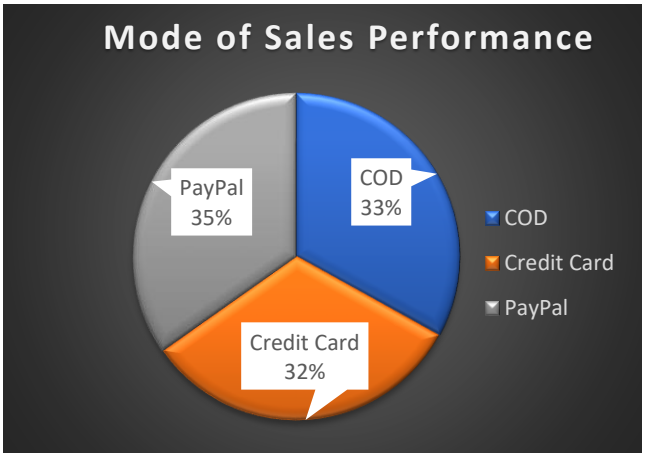
STRATEGIC INSIGHTS:

- Sales show clear seasonal patterns (Q4 & March–April peak).
- Moderate discounting (14–16%) supports higher revenue months.
- 2024 was the most stable and strongest year overall.
- 2025 shows volatility and possible incomplete Q4 data.
- November 2024 is the best-performing month in the dataset.

💼 BUSINESS RECOMMENDATIONS:

- ✓ Increase marketing and inventory before November and March–April.
- ✓ Maintain discount strategy around 15% for peak conversion.
- ✓ Investigate 2025 mid-year dip for operational causes.
- ✓ Monitor incomplete month data before strategic decisions.

3)Mode of Sale Performance:



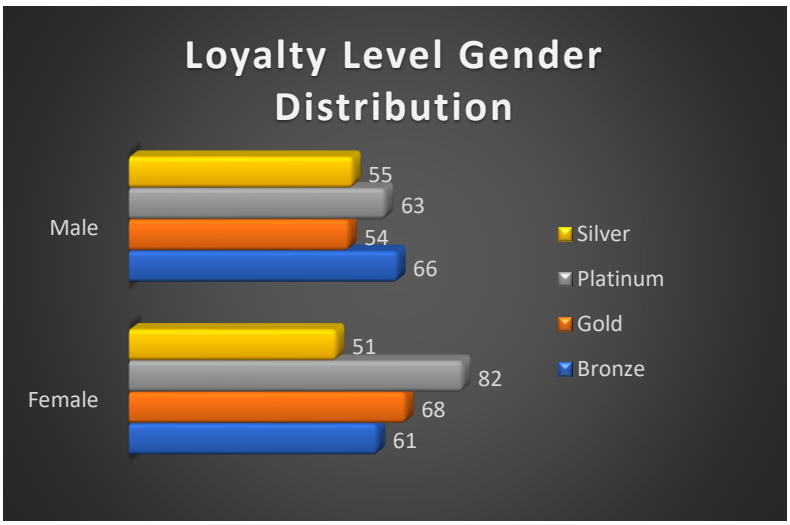
STRATEGIC INSIGHTS:

- ➢ Payment preference is evenly distributed (no dominant monopoly).
- ➢ Digital payments (PayPal/Credit Card) generate higher average spending than COD.
- ➢ Credit card users represent higher-value customers.

💼 **BUSINESS RECOMMENDATIONS:**

- ✓ Promote credit card cashback offers to increase high-value transactions.
- ✓ Encourage COD customers to shift to digital payments via discounts.
- ✓ Strengthen PayPal promotions since it drives both volume and revenue.

4)Loyalty Level Gender Distribution:



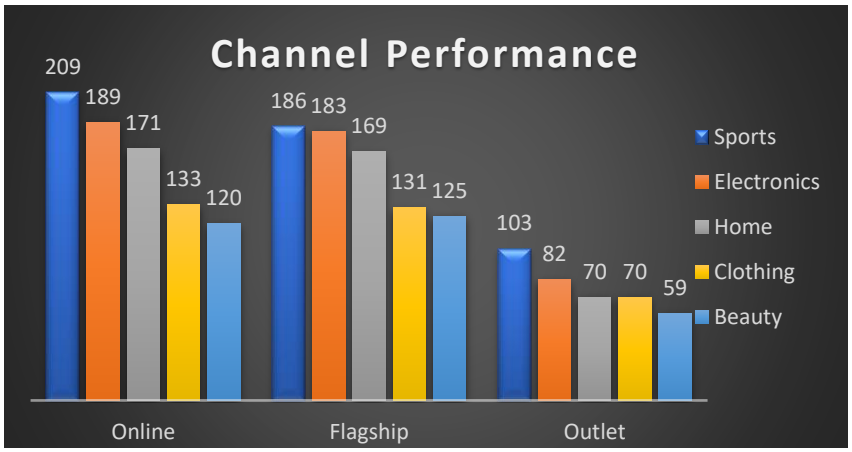
STRATEGIC INSIGHTS:

- Female customers are more represented in higher loyalty tiers (Gold & Platinum).
- Male customers are slightly more concentrated in Bronze level.
- Platinum being the largest tier indicates strong customer retention overall.
- Silver has the lowest count — possible upgrade opportunity segment.

💼 **BUSINESS RECOMMENDATIONS:**

- ✓ Target male Bronze customers with loyalty upgrade campaigns.
- ✓ Reward female Platinum customers with exclusive perks to maintain retention.
- ✓ Create promotional campaigns to convert Silver customers into Gold tier.
- ✓ Analyze spending behavior by gender to confirm loyalty-spending correlation.

5) Channel Performance Analysis:



Online-41%, Flagship-40%, Outlet-19%

STRATEGIC INSIGHTS:

- Online channel drives overall sales growth.
- Flagship stores are competitive and strong in premium categories.
- Outlet stores underperform significantly.
- Sports & Electronics are top revenue-driving categories.
- Beauty performs better in physical premium stores.

💼 **BUSINESS RECOMMENDATIONS:**

- ✓ Invest further in Online marketing & logistics.
- ✓ Strengthen Beauty promotions in Flagship stores.
- ✓ Revamp Outlet strategy (pricing, assortment, or promotions).
- ✓ Expand Sports inventory across all channels.

KPI Metrics:

- ❖ Total Revenue: ₹21,74,394.70
- ❖ Average Order Value: ₹1087.20
- ❖ Highest Revenue Month: November 2024
- ❖ Top Region: North

- Top Category: Sports

Conclusion:

The E-Commerce sales analysis demonstrates a well-performing but strategically improvable business structure. The North region leads overall sales performance, while the West region requires targeted marketing intervention. Sports and Electronics emerge as the most revenue-generating categories across all store types. Online and Flagship channels contribute nearly equal sales shares, with Outlet stores significantly underperforming. Seasonal patterns indicate strong sales peaks during Q4 and early Q2, supported by moderate discount strategies around 15%. Loyalty analysis shows stronger premium-tier participation among female customers, highlighting retention opportunities among male and mid-tier segments. Overall, the business demonstrates stable growth potential, with opportunities for channel optimization, regional expansion, and strategic promotional planning.